SUPPLY CHAIN VISIBILITY SYSTEM WITH OPTIMIZATION ANALYTICS

Power BI Project Documentation

Data Modelling & KPI Foundation | Interactive Dashboard | DAX Analytics

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Tools Used: CSV Dataset | Power BI Desktop | Power Query | DAX

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Batch : 03

SUPPLY CHAIN VISIBILITY SYSTEM WITH OPTIMIZATION ANALYTICS

1. Introduction

The **Supply Chain Visibility System with Optimization Analytics** is a Power BI-based data analytics project developed to analyze and improve supply chain operations. The project provides a centralized view of important business activities such as **sales, inventory, orders, suppliers, warehouses, delivery performance, transportation costs, and profitability**.

The project is developed in multiple milestones. Each milestone focuses on a different area of supply chain analysis. The dashboards and analytical reports help users understand operational performance, identify variations, compare warehouses and suppliers, analyze transportation activities, and support better business decisions.

The project uses **Power BI, Power Query, DAX, and Excel/CSV data** to transform raw supply chain data into interactive dashboards and analytical reports.

The four milestones cover:

- **Milestone 1:** Data Preparation, Data Modeling and Basic Supply Chain Analytics
- **Milestone 2:** Inventory and Delivery Analytics
- **Milestone 3:** Supplier and Transportation Analytics
- **Milestone 4:** Dashboard Performance and Optimization Analytics

2. Problem Statement

Supply chain operations generate large amounts of data related to **sales, inventory, orders, suppliers, warehouses, demand, delivery time, and transportation**. When this information is maintained as raw data or analyzed separately, it becomes difficult to identify performance problems and make timely decisions.

The main problems addressed by this project are:

- Difficulty in monitoring overall supply chain performance.
- Difficulty in analyzing sales, profit, and inventory together.
- Lack of clear visibility into warehouse performance.
- Difficulty in identifying inventory differences between supply and demand.
- Difficulty in comparing supplier delivery performance.
- Lack of visibility into transportation costs and distances.
- Difficulty in analyzing routes, carriers, and transportation modes.
- Difficulty in identifying profitable and low-performing warehouses.
- Lack of a centralized interactive dashboard for management.
- Manual analysis of large datasets can be time-consuming and error-prone.

Therefore, this project aims to develop an **interactive Power BI-based Supply Chain Visibility System** that converts raw data into meaningful dashboards, KPIs, trends, comparisons, and optimization-oriented insights.

3. Tools and Technologies Used

Tool / Technology	Purpose
Microsoft Power BI	Dashboard creation, data visualization and interactive reporting
Power Query	Data cleaning, transformation and preparation
DAX	Creating calculated columns and analytical measures
Microsoft Excel	Dataset preparation and data management
CSV	Source data format
Data Modeling	Creating relationships between fact and dimension tables
Star Schema	Organizing the data model for efficient analysis
Power BI Visuals	Charts, cards, slicers, matrices and analytical dashboards

Power BI Visuals Used

The project uses different visualizations according to the requirements of each milestone, including:

- KPI Cards
- Clustered Column Charts
- Clustered Bar Charts
- Line Charts
- Donut Charts
- Pie Charts
- Treemaps
- Ribbon Charts
- Scatter Charts
- Matrix
- Slicers

4. Dataset

The main dataset used in the project contains supply chain information for **2024** and includes daily SKU-level records across warehouses and suppliers.

The main dataset contains **91,250 records and 15 fields**.

The dataset includes information related to:

Dataset Category	Description
Date	Date of the supply chain transaction/record
SKU	Stock Keeping Unit/product identifier
Warehouse	Warehouse associated with the record
Supplier	Supplier associated with the record
Region	Geographical region
Units Sold	Quantity of products sold
Unit Price	Selling price of the product
Unit Cost	Cost of the product
Inventory Level	Available inventory level
Order Quantity	Quantity ordered
Supplier Lead Time	Supplier delivery/lead time
Demand Forecast	Forecasted demand
Stockout Flag	Indicates stockout status
Sales Amount	Calculated sales value
COGS / Inventory-related values	Calculated cost and inventory metrics

The dataset was cleaned and transformed using **Power Query** before being used for analysis.

5. Data Model

The project follows a **star-schema-based data model**.

Main Fact Table

FactSupplyChain

Contains transactional supply chain information such as:

- Sales
- Units Sold
- Inventory
- Order Quantity
- Demand Forecast
- Supplier Lead Time
- Stockout Flag
- Unit Price
- Unit Cost

Dimension Tables

- **DimDate**
- **DimSKU**
- **DimSupplier**
- **DimWarehouse**

These dimension tables are connected to the main fact table using appropriate relationships.

For Milestone 3, an additional transportation table was created:

FactTransportation

It contains transportation-related information such as:

- Date
- Supplier ID
- Warehouse ID
- Route ID
- Carrier ID
- Transport Mode
- Distance
- Transportation Cost
- Delivery Days

The transportation records were added as **sample/synthetic records** because the original dataset did not contain fields such as Carrier, Route, Transportation Cost, Distance and Transport Mode.

MILESTONE 1

Data Modelling & KPI Foundation Supply Chain Visibility System with Optimization Analytics

1. Explanation of Milestone 1

Supply chains involve the continuous movement of goods from suppliers to warehouses and finally to customers. Organizations need clear visibility into sales, inventory, supplier performance, and stock availability to operate efficiently.

This milestone analyzes the supply-chain dataset supplied as a CSV export using Power BI and converts the raw transactional data into a **clean data model, a set of KPIs, and an interactive dashboard** that supports quick, data-driven decisions.

2. Problem Statement

Supply-chain teams often work with raw transactional exports containing sales, cost, inventory, supplier, and warehouse data. Without a unified model, it is difficult to answer important questions such as:

- Which warehouse is overstocked?
- Which supplier is consistently late?
- Which products are not selling?
- How is inventory distributed across warehouses?
- What is the overall sales and profitability performance?

This project addresses this gap by creating a **single connected Power BI model and dashboard** that brings the information together in one place.

3. Tools & Technologies

Tool	Purpose
CSV File (supply_chain_dataset1.csv)	Source format for the raw supply-chain dataset
Power Query	Data cleaning, transformation and shaping
Power BI Desktop	Data modelling, DAX and dashboard development
DAX	KPI calculations and measures

Power BI Desktop was selected as the core tool because it combines **data ingestion, Power Query, data modelling, relationships, DAX calculations, and visualization** in one application.

4. Dataset Description

The project uses a single fact table named **FactSupplyChain**, containing daily transaction-level records across products, suppliers, warehouses, and regions.

Field	Description
Date	Transaction date
SKU_ID	Unique product identifier
Supplier_ID	Supplier identifier
Warehouse_ID	Warehouse identifier
Region	Sales region
Sales / COGS	Revenue and cost of goods sold
Inventory_Level / Inventory_Value	Stock quantity and its value
Order_Quantity / Reorder_Point	Ordering and replenishment thresholds
Demand_Forecast	Forecasted demand
Supplier_Lead_Time	Days taken by supplier to deliver

Each row in FactSupplyChain represents one SKU at one warehouse on one date, together with sales, cost, and inventory information. The grain is:

Date × SKU × Warehouse × Supplier

This level of detail supports both high-level KPI cards and detailed product-level analysis.

5. Data Cleaning Using Power Query

The raw CSV data was imported into Power BI using:

Home → Get Data → Text/CSV → Transform Data

The following cleaning activities were performed in Power Query:

5.1 Column Review

Columns were reviewed and renamed where necessary for clarity.

5.2 Data Typing

Appropriate data types were assigned:

- Date → Date
- Quantity fields → Whole Number
- Money fields → Decimal Number
- ID fields → Text

5.3 Duplicate Removal

Duplicate rows were removed where uniqueness was expected.

5.4 Blank Row Removal

Blank records left over from the CSV export were removed.

5.5 Missing Value Handling

Occasional blank values were handled using **Replace Values** where required.

5.6 Apply Changes

After cleaning, **Close & Apply** was used to load the cleaned data into the Power BI model.

Cleaning the data before modelling is important because incorrect data types can affect DAX calculations, while duplicate keys can cause problems with one-to-many relationships.

6. Data Modelling – Star Schema

A **star schema** was created to keep the Power BI model simple, efficient, and easy to filter.

The model contains:

- **FactSupplyChain**
- **DimDate**
- **DimSKU**
- **DimSupplier**
- **DimWarehouse**

The dimension tables are on the **"1" side**, while FactSupplyChain is on the **"many" side** of the relationships.

Model Structure

DimDate → FactSupplyChain
DimSKU → FactSupplyChain
DimSupplier → FactSupplyChain
DimWarehouse → FactSupplyChain

This structure allows a slicer selection such as a particular warehouse to filter the related fact-table records and update the dashboard visuals.

7. Dimension Tables

7.1 DimDate

DimDate is the date dimension used for time-based analysis and time intelligence.

It helps analyze:

- Sales by date
- Sales by month
- Sales by year
- Inventory trends

DAX

```
DimDate =
CALENDAR(
   MIN(FactSupplyChain[Date]),
   MAX(FactSupplyChain[Date])
)
```

Relationship:

DimDate[Date] 1 → * FactSupplyChain[Date]

7.2 DimSKU

DimSKU is the product dimension containing information about different products or SKUs.

It helps analyze:

- Product sales
- Product inventory
- Product-level KPIs

Example measure:

Average Inventory =
AVERAGE(FactSupplyChain[Inventory_Level])

Relationship:

DimSKU[SKU_ID] 1 → * FactSupplyChain[SKU_ID]

7.3 DimSupplier

DimSupplier stores information about suppliers.

It helps compare supplier performance and supplier lead time.

Example DAX:

Average Supplier Lead Time =
AVERAGE(FactSupplyChain[Supplier_Lead_Time])

Relationship:

DimSupplier[Supplier_ID] 1 → * FactSupplyChain[Supplier_ID]

7.4 DimWarehouse

DimWarehouse stores information about different warehouses.

It helps compare:

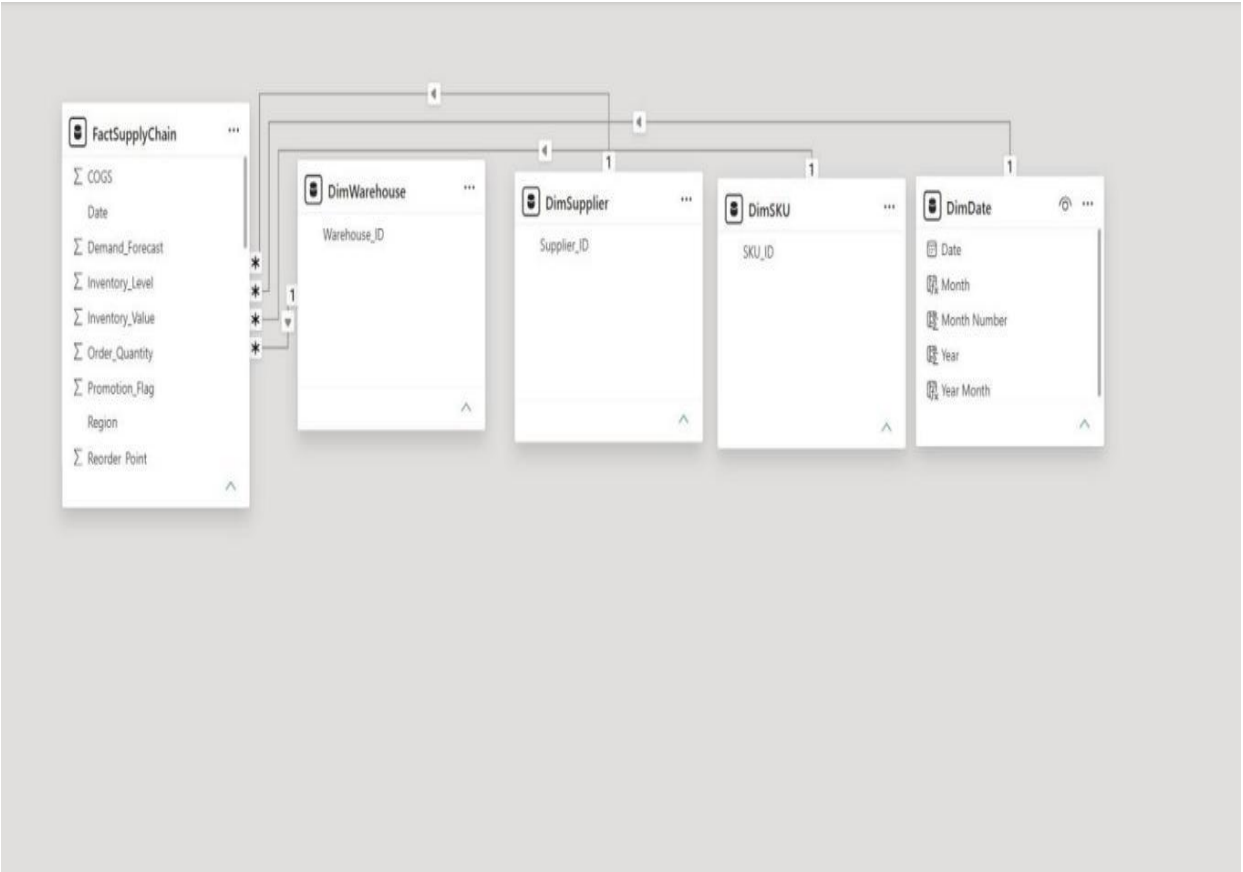
- Inventory levels
- Inventory value
- Warehouse performance

Example DAX:

Total Inventory Value =
SUM(FactSupplyChain[Inventory_Value])

Relationship:

DimWarehouse[Warehouse_ID] 1 → * FactSupplyChain[Warehouse_ID]



8. DAX Measures

DAX (**Data Analysis Expressions**) was used to convert raw columns into business-ready calculations.

The key measures created were:

Measure	Purpose
Total Sales	Calculates total sales/revenue
Total COGS	Calculates cost of goods sold
Total Units Sold	Calculates total quantity
Total Inventory Value	Calculates inventory value
Average Inventory Level	Calculates average stock level
Profit	Calculates Sales − COGS
Profit Margin %	Calculates profitability percentage
Average Supplier Lead Time	Calculates average supplier delivery time
Stockout Rate	Measures stockout frequency
Inventory Turnover	Measures inventory movement

Example formulas from the milestone include:

1.Total Sales = SUM(FactSupplyChain[Sales_Amount])
2.Total COGS = SUM(FactSupplyChain[COGS])
3.Profit = [Total Sales] - [Total COGS]
4.Profit Margin % = DIVIDE([Profit], [Total Sales], 0)
5.Average Inventory Level = AVERAGE(FactSupplyChain[Inventory_Level])
6.Average Supplier Lead Time = AVERAGE(FactSupplyChain[Supplier_Lead_Time])

9. Key Performance Indicators (KPIs)

The DAX measures were displayed as **KPI cards** so that important information could be understood quickly.

KPI	Meaning
Total Sales	Total revenue generated
Total COGS	Cost of goods sold
Total Units Sold	Total quantity sold
Total Inventory Value	Value of stock held
Profit / Profit Margin	Business profitability
Stockout Rate	Frequency of products going out of stock
Supplier Lead Time	Average supplier delivery time

10. Dashboard Design

An interactive Power BI report page titled:

Supply Chain Visibility System – Optimization Analytics was created using:

- KPI cards
- Charts
- Slicers
- Interactive filtering

The dashboard allows users to analyze information by:

- Warehouse
- SKU
- Supplier
- Time period

11. KPI Cards

The dashboard contains the following KPI cards:

- **Total COGS**
- **Total Units Sold**
- **Total Inventory Value**
- **Profit**
- **Total Sales**

These provide a quick summary of the overall supply-chain performance.

12. Dashboard Charts

The following charts were created:

12.1 Total Inventory Value by Region

Visual: Clustered Column Chart

Used to compare inventory value across:

- North
- South
- East
- West

12.2 Total Inventory Value by Warehouse

Visual: Donut Chart

Used to understand warehouse-wise stock distribution.

12.3 Average Inventory Level by SKU

Visual: Horizontal Bar Chart

Used to identify products with relatively high or low inventory levels.

12.4 Warehouse Inventory Level

Visual: Pie Chart

Used to highlight warehouse inventory concentration.

12.5 Sales and Inventory Value by Year Month

Visual: Line Chart

Used to analyze monthly sales and inventory trends.

13. Interactive Slicers

The following slicers were added:

Slicer	Purpose
Year Month	Filter the report by time
Warehouse_ID	Filter by warehouse
SKU_ID	Filter by product
Supplier_ID	Filter by supplier

Because the visuals are connected through the star-schema relationships, selecting one slicer automatically updates the related KPI cards and charts.

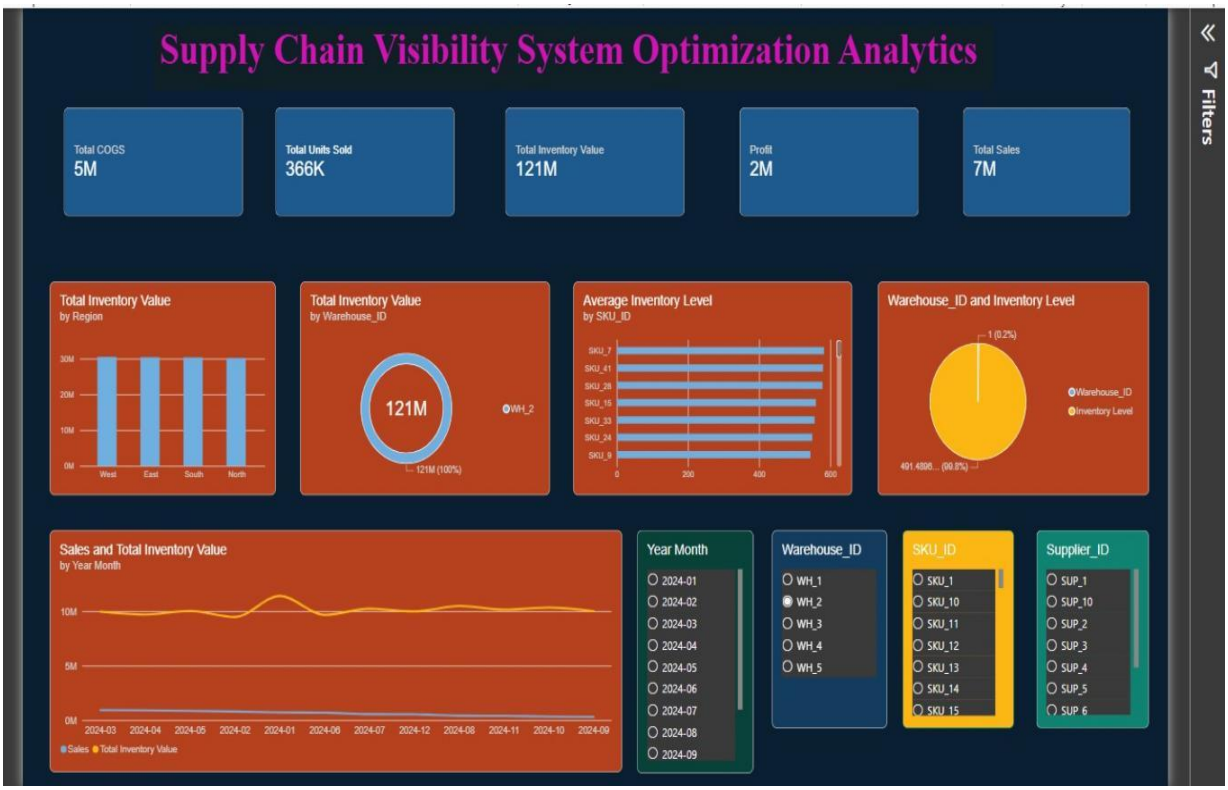
The dashboard helps identify several important supply-chain situations:

- **High demand + low inventory** → may indicate the need to reorder stock sooner.
- **Low demand + high inventory** → may indicate overstocking.
- **High Stockout Rate** → may indicate a need for higher safety stock or improved ordering.
- **Warehouse comparison** → helps identify locations with excess or insufficient inventory.
- **Slow-moving SKUs** → products with low turnover and high inventory value can be investigated.
- **Supplier lead time** → helps identify suppliers that may require closer monitoring.

16. Project Workflow

The overall Milestone 1 workflow can be explained as:

Raw CSV Dataset
↓
Power Query Data Cleaning
↓
Data Transformation
↓
Star Schema Data Modelling
↓
Relationships
↓
DAX Measures
↓
KPI Cards
↓
Charts & Slicers
↓
Interactive Supply Chain Dashboard



MILESTONE 2

Inventory & Delivery Analytics - Inventory Gap Analysis | Supplier Delivery Performance | Trend & Variance Analysis

1. Introduction

Milestone 2 focuses on analyzing the **inventory and delivery performance** of the Supply Chain Visibility System.

After completing data cleaning, data modelling, relationships, and basic KPI development in Milestone 1, this milestone uses Power BI to transform the supply-chain data into meaningful analytical reports that go beyond the foundational KPIs and support operational decision-making.

The main purpose of this milestone is to help management understand:

- Whether available inventory is sufficient compared with forecasted demand.
- Which warehouses have higher or lower inventory gaps.
- Which products/SKUs have higher average inventory levels.
- Which suppliers have higher delivery lead times.
- How delivery performance varies across regions.
- How sales and inventory change over time.
- How sales compare with COGS across warehouses.

The dashboard uses **interactive charts, KPI cards, and slicers** so users can analyze the data by warehouse, product, supplier, region, and time period without requiring separate reports for each view.

2. Problem Statement

The main objectives are:

- Develop inventory performance reports.
- Create delivery performance dashboards.
- Implement regional-level analysis.
- Implement product/SKU-level analysis.
- Perform trend analysis.
- Perform variance analysis.
- Create interactive filtering using slicers.
- Provide KPI-based monitoring of delivery performance.

3. Inventory Performance Analysis

Inventory is one of the most important components of supply-chain management. Maintaining the correct inventory level is important because both excess inventory and insufficient inventory can create business problems.

Excess inventory

Excess inventory can increase:

- Storage costs
- Holding costs
- Inventory management requirements

Insufficient inventory

Insufficient inventory can make it difficult to satisfy expected demand. Therefore, comparing inventory with demand helps identify potential inventory gaps.

The dataset contains **Inventory_Level** and **Demand_Forecast** information, which are used as the foundation for this analysis.

4. Inventory vs Demand

To compare available inventory with forecasted demand, the following DAX measure was created:

Inventory vs Demand = SUM(FactSupplyChain[Inventory_Level]) -
SUM(FactSupplyChain[Demand_Forecast])

Formula

Inventory vs Demand = Inventory − Forecasted Demand

Working

The measure calculates:

1. Total Inventory
2. Total Forecasted Demand
3. Difference between Inventory and Forecasted Demand

Interpretation

Result	Interpretation
Positive	Inventory is higher than forecasted demand
Negative	Inventory is lower than forecasted demand
Zero	Inventory and forecasted demand are equal

This measure provides the foundation for inventory gap analysis.

5. Inventory Gap Percentage

A percentage-based measure was created to make inventory differences easier to compare between warehouses.

DAX Measure

```
Inventory Gap % =
DIVIDE(
    [Inventory vs Demand],
    SUM(FactSupplyChain[Demand_Forecast]),
    0
)
```

Formula

Inventory Gap % = (Inventory − Demand) / Demand

The DIVIDE() function is used because it safely handles situations where demand is zero and prevents calculation errors in dashboard visuals.

Business Interpretation

For example:

- A **positive 20%** inventory gap indicates inventory is 20% higher than forecasted demand.
- A **negative percentage** indicates inventory is below forecasted demand.

A negative inventory gap may indicate a possible inventory shortage and requires further investigation.

6. Inventory Gap by Warehouse

A **Clustered Column Chart** was created to visualize inventory gap percentage across warehouses.

Chart Property	Field
Visual	Clustered Column Chart
X-axis	Warehouse
Y-axis	Inventory Gap %
Title	Inventory Gap % by Warehouse

Purpose

This chart helps compare inventory positions between warehouses.

It helps answer:

- Which warehouse has a relatively high inventory gap?
- Which warehouse has inventory closer to forecasted demand?
- Which warehouses may require further inventory investigation?

The warehouse-level comparison helps planners prioritize locations for further review.

7. Product-Level Inventory Analysis

Inventory performance was also analyzed at the **SKU/product level**.

SKU stands for **Stock Keeping Unit**. Each SKU represents a specific product or product identifier used for inventory tracking.

The dataset contains SKU-level information, allowing inventory analysis to be performed for individual products rather than only at warehouse or company level.

7.1 Average Inventory Level

The following DAX measure was used:

Average Inventory Level = AVERAGE(FactSupplyChain[Inventory_Level])

Purpose

Instead of simply looking at total inventory, the average inventory level shows the typical inventory level recorded for each SKU over the available data.

This is useful because the dataset contains inventory observations across different dates, and simply summing repeated inventory snapshots could overstate the stock position.

8. Average Inventory Level by SKU

A **Clustered Bar Chart** was created.

Chart Property	Field
Visual	Clustered Bar Chart
Y-axis	SKU
X-axis	Average Inventory Level
Title	Average Inventory Level by SKU

The chart was sorted in descending order.

Purpose

This visualization helps identify products with relatively high average inventory levels.

It supports:

- Inventory monitoring
- Product-level stock analysis
- Storage planning
- Identifying products requiring further investigation

9. Delivery Performance Analysis

Delivery performance is another important part of supply-chain management.

The project dataset contains **Supplier_Lead_Time**, which represents delivery-related time associated with suppliers.

In this project, delivery performance is analyzed using **Average Supplier Lead Time** rather than an invented delivery-status field, keeping the analysis based on the available data.

10. Average Supplier Lead Time

The following DAX measure was created:

Average Supplier Lead Time = AVERAGE(FactSupplyChain[Supplier_Lead_Time])

Purpose

This measure calculates the average supplier lead time from the available supply-chain records.

A lower lead time generally represents a shorter supplier delivery cycle, while a higher lead time indicates a longer delivery cycle.

Importance

Monitoring supplier lead time helps organizations:

- Compare suppliers.
- Understand delivery-cycle differences.
- Identify suppliers requiring closer monitoring.
- Support supplier-management decisions.
- Improve supply-chain planning.

11. Average Supplier Lead Time by Supplier

A **Clustered Bar Chart** was created to compare suppliers.

Chart Property	Field
Visual	Clustered Bar Chart
Y-axis	Supplier
X-axis	Average Supplier Lead Time
Title	Average Supplier Lead Time by Supplier

The chart was sorted in descending order.

Purpose

The chart provides a supplier-level comparison of average lead time.

A supplier toward the higher end of the chart has a relatively higher average lead time and may require additional monitoring.

12. Delivery Performance KPI

A **Card visual** was added to provide a high-level view of delivery performance.

Property	Value
Visual	Card
Measure	Average Supplier Lead Time
Title	Average Delivery Lead Time

Purpose

The KPI card provides a single summary value that can be understood quickly without examining the entire supplier chart.

This is useful for management-level dashboard viewing because the overall delivery lead-time metric is immediately visible.

13. Regional Delivery Performance

Delivery performance was further analyzed based on **Region**.

A **Clustered Column Chart** was created.

Chart Property	Field
Visual	Clustered Column Chart
X-axis	Region
Y-axis	Average Supplier Lead Time
Title	Average Supplier Lead Time by Region

Purpose

This chart helps compare average supplier lead time between different regions.

It allows users to identify regional differences in delivery lead time and investigate whether particular regions require additional attention.

For example, a region with consistently higher lead times might require closer supplier monitoring or consideration of alternative suppliers.

14. Trend Analysis

Trend analysis is used to understand how important supply-chain measures change over time.

For this project, **Sales and Inventory** were analyzed together using a Line Chart.

Chart Property	Field
Visual	Line Chart
X-axis	Year Month
Series 1	Total Sales
Series 2	Total Inventory Value
Title	Sales & Inventory Trend

The Year Month field was arranged chronologically using the corresponding sorting column so that the trend appears in true calendar order.

14.1 Total Sales

The existing measure was used:

```
Total Sales =
SUM(FactSupplyChain[Sales_Amount])
```

Sales amount was derived from **Units Sold × Unit Price** during data preparation.

14.2 Total Inventory Value

The inventory value measure was used to display the inventory-value series in the trend chart.

The calculated inventory-value field was prepared during the data transformation stage.

Purpose of Trend Analysis

The line chart helps users observe:

- Changes in sales over time.
- Changes in inventory value over time.
- Periods where sales and inventory move differently.
- General patterns in supply-chain activity.

This provides a time-based view instead of looking only at a single overall value.

15. Variance Analysis

Variance analysis was introduced to compare **Sales and COGS**. **COGS** means **Cost of Goods Sold**. It represents the cost associated with the goods sold. The project includes both Sales and COGS as important financial measures.

16. Sales vs COGS Variance

The following DAX measure was created:

Sales vs COGS Variance = [Total Sales] - [Total COGS]

Working

The measure subtracts Total COGS from Total Sales. It provides the difference between:

Revenue generated from Sales − Cost associated with sold goods

This gives a simple view of the financial difference between sales and COGS.

17. Sales vs COGS Variance by Warehouse

A **Clustered Column Chart** was created.

Chart Property	Field
Visual	Clustered Column Chart
X-axis	Warehouse
Y-axis	Sales vs COGS Variance
Title	Sales vs COGS Variance by Warehouse

Purpose

The chart allows users to compare the sales-to-COGS difference across warehouses.

It can help management understand how the financial difference varies between warehouse operations and identify warehouses that may require further investigation.

18. Interactive Slicers

Interactive slicers were added to make the dashboard dynamic.

Slicer	Field	Style
Period	Year Month	List
Warehouse	Warehouse	List
Product	SKU	List
Supplier	Supplier	List

Purpose of Slicers

Slicers allow users to filter the dashboard without manually changing each visual.

For example:

- Selecting a particular **Warehouse** filters the analysis to that warehouse.
- Selecting a particular **SKU** allows product-specific analysis.
- Selecting a **Supplier** focuses the delivery analysis on that supplier.
- Selecting a **Year Month** filters the dashboard for the selected period.

This makes the report interactive and easier to explore.

19. DAX Measures Created

Measure	Formula
Inventory vs Demand	SUM(Inventory_Level) − SUM(Demand_Forecast)
Inventory Gap %	DIVIDE([Inventory vs Demand], SUM(Demand_Forecast), 0)
Average Inventory Level	AVERAGE(Inventory_Level)
Average Supplier Lead Time	AVERAGE(Supplier_Lead_Time)
Sales vs COGS Variance	[Total Sales] − [Total COGS]

20. Dashboard Components

The following components were completed as part of Milestone 2:

No.	Dashboard Component	Analysis Performed
1	Inventory Gap % by Warehouse	Inventory vs Forecasted Demand
2	Average Inventory Level by SKU	Product-Level Inventory
3	Average Supplier Lead Time by Supplier	Supplier Delivery Analysis
4	Average Delivery Lead Time	Overall Delivery KPI
5	Average Supplier Lead Time by Region	Regional Delivery Analysis
6	Sales & Inventory Trend	Time-Based Trend Analysis
7	Sales vs COGS Variance by Warehouse	Warehouse-Level Financial Variance
8	Year Month Slicer	Period Filtering
9	Warehouse Slicer	Warehouse Filtering
10	SKU Slicer	Product Filtering
11	Supplier Slicer	Supplier Filtering

20. Overall Analytical Flow of Milestone 2

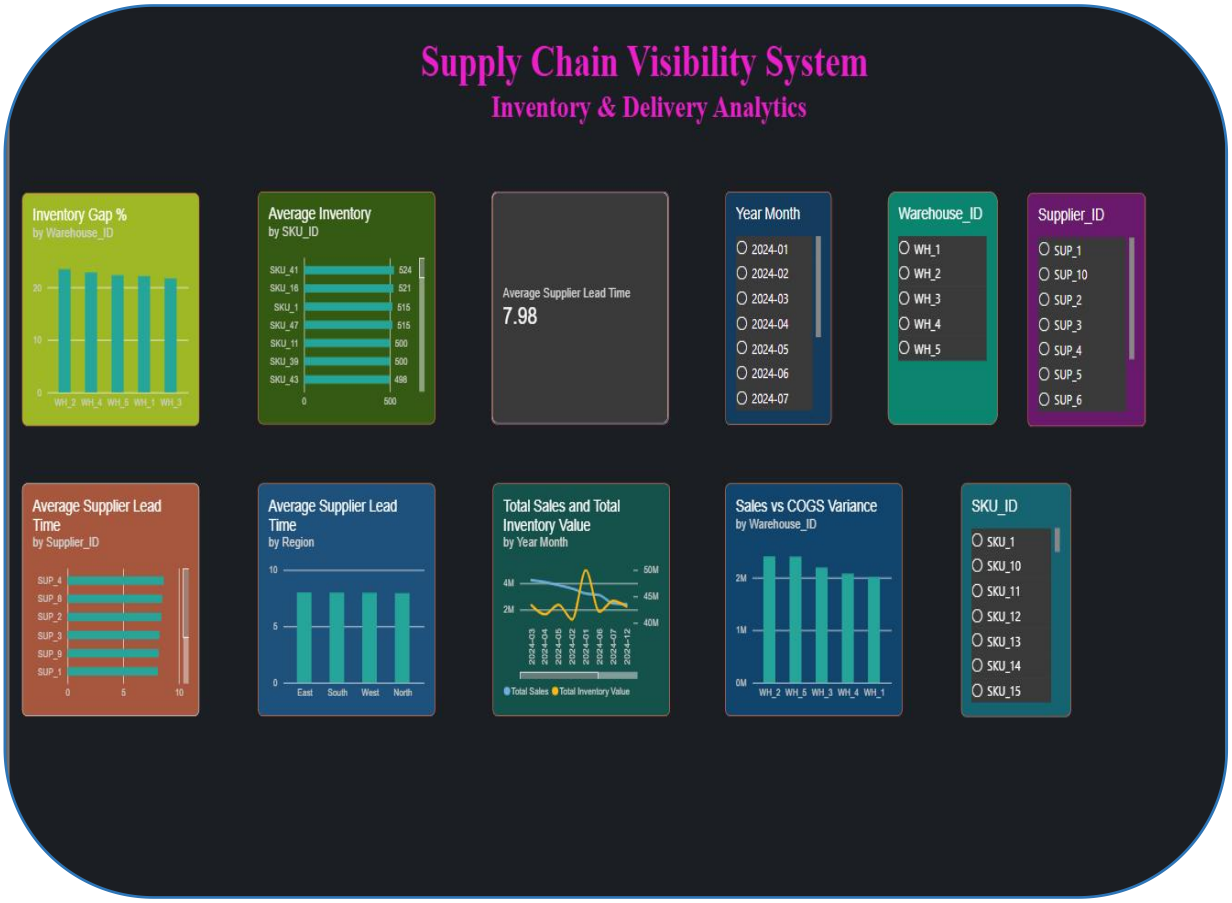
The analytical process implemented in this milestone can be represented as:

Supply Chain Data
↓
Inventory & Delivery Measures
↓
Inventory Gap Analysis
↓
Product & Warehouse Analysis
↓
Supplier & Regional Delivery Analysis
↓
Trend Analysis
↓
Sales vs COGS Variance Analysis
↓
Interactive Slicers
↓
Inventory & Delivery Analytics Dashboard

The resulting dashboard provides a centralized view of:

- Inventory
- Supplier delivery lead time
- Regional performance
- Product-level inventory
- Time trends
- Warehouse-level variance analysis

It builds directly on the data model and KPI foundation established in Milestone 1.



Milestone 2 extends the foundation created in Milestone 1 by focusing on **inventory sufficiency, product-level inventory, supplier delivery performance, regional delivery differences, sales and inventory trends, and warehouse-level financial variance**.

The use of **DAX measures, interactive charts, KPI cards, and slicers** makes the dashboard useful for operational analysis and helps users identify areas that require further investigation.

MILESTONE 3

Supplier & Transportation Analytics - Supplier Scorecards | Transportation Cost | Route & Carrier Analysis | Advanced KPIs | Interactive Dashboard

1. Introduction

Milestone 3 extends the Supply Chain Visibility System from inventory and delivery analytics into **supplier and transportation analytics**.

Milestone 1 focused on establishing a clean relational data model and core DAX KPIs. Milestone 2 added inventory-gap analysis, product-level inventory analysis, supplier lead-time analysis, regional delivery analysis, trend analysis and variance analysis.

Milestone 3 shifts the analytical focus toward **how goods physically move between suppliers, carriers and warehouses**.

The main questions addressed are:

- Which suppliers are efficient?
- Which routes have higher transportation costs?
- Which carriers account for higher transportation expenditure?
- Which transportation modes are more expensive?
- Which transportation modes provide faster delivery?
- Which shipments have unusually high costs compared with their distance?

The original dataset used in Milestones 1 and 2 contains **Supplier Lead Time**, but it does not contain dedicated **Carrier, Route, Transport Mode, Distance or Transportation Cost** fields.

Therefore, a separate **FactTransportation** table was added specifically for Milestone 3. These transportation records are **synthetic/sample records** created to support the transportation analysis and are kept separate from the original FactSupplyChain table.

2. Problem Statement

The main objectives of Milestone 3 are:

- Build supplier performance scorecards.
- Develop transportation cost dashboards.
- Create route and carrier performance analysis.
- Compare transportation modes such as Road, Rail and Air.
- Implement advanced transportation KPIs using DAX.
- Analyze delivery duration and transportation efficiency together.
- Provide interactive filtering using supplier, warehouse, period and transportation-mode slicers.
- Integrate transportation analytics without creating a direct fact-to-fact relationship.

3. Transportation Data Added

A separate **FactTransportation** table was created as a purpose-built fact table for logistics analysis. It contains only the fields required for transportation analysis and connects to the existing dimension tables.

Field	Description
Date	Transportation transaction date
Supplier_ID	Supplier associated with the transportation record
Warehouse_ID	Destination warehouse
Route_ID	Transportation route identifier
Carrier_ID	Logistics provider/carrier identifier
Transport_Mode	Mode of transport such as Road, Rail and Air
Distance_KM	Transportation distance in kilometres
Transportation_Cost	Cost incurred for transportation
Delivery_Days	Delivery duration in days

4. FactTransportation Table

The transportation data was kept separate from FactSupplyChain for three main reasons.

4.1 Different Data Grain

Supply-chain transactions and transportation events do not necessarily have the same grain.

One supply-chain transaction can correspond to zero, one, or multiple transportation legs.

4.2 Protecting the Existing Model

The validated Milestone 1 and Milestone 2 model was not modified.

This reduces the risk of breaking the existing relationships and calculations.

4.3 Realistic Data Modelling

In real organizations, transportation information often comes from a separate **Transportation Management System (TMS)** or carrier data source.

Therefore, maintaining a separate transportation fact table makes the model more realistic.

5. Data Modelling

FactTransportation was connected to the existing dimension tables so that transportation visuals respond to the same filters used throughout the project.

The relationships are:

Relationship	Cardinality	Purpose
DimSupplier[Supplier_ID] → FactTransportation[Supplier_ID]	1 → *	Supplier filtering
DimWarehouse[Warehouse_ID] → FactTransportation[Warehouse_ID]	1 → *	Warehouse filtering
DimDate[Date] → FactTransportation[Date]	1 → *	Time-based filtering

All relationships use **single-direction filtering from dimension to fact**.

Important Modelling Decision

FactTransportation is **not directly connected to** FactSupplyChain.

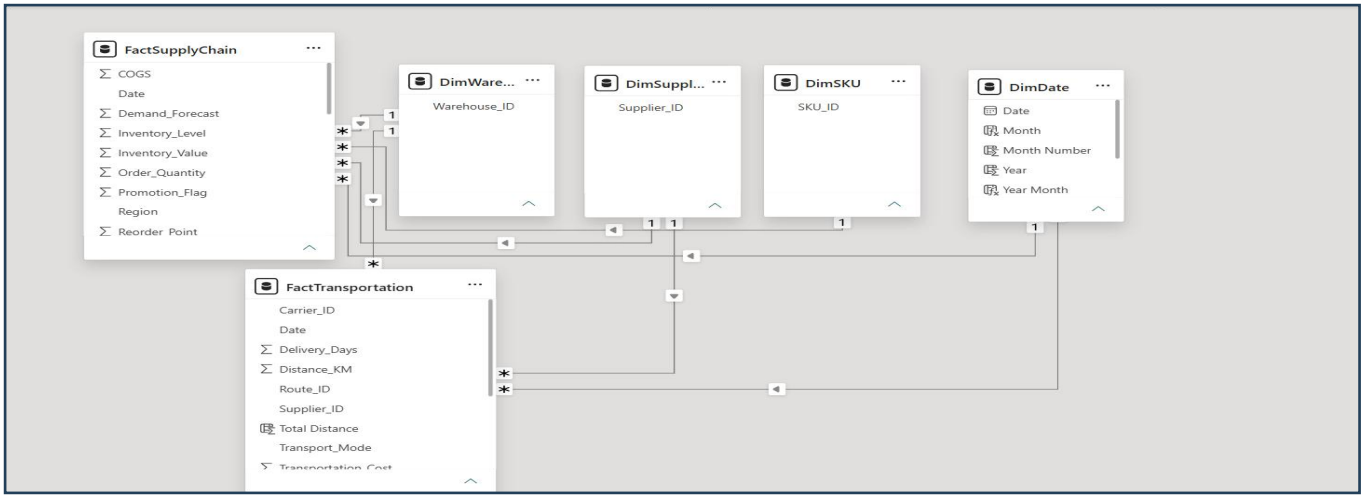
Instead, the shared dimensions act as the bridge:

DimSupplier → FactSupplyChain
DimSupplier → FactTransportation

DimWarehouse → FactSupplyChain
DimWarehouse → FactTransportation

DimDate → FactSupplyChain
DimDate → FactTransportation

This avoids ambiguous filter paths and maintains the star-schema structure.



6. Transportation KPI Measures

Five main DAX measures were created.

6.1 Total Transportation Cost

Total Transportation Cost = SUM(FactTransportation[Transportation_Cost])

Purpose

Calculates the total transportation expenditure for the selected filter context.

It can be used to analyze:

- Overall transportation spending
- Supplier transportation cost
- Route cost
- Carrier cost
- Transportation-mode cost

6.2 Average Transportation Cost

Average Transportation Cost = AVERAGE(FactTransportation[Transportation_Cost])

Purpose

Calculates the average transportation cost per transportation record.

This answers:

“What is the typical transportation cost?”

This measure is useful alongside Total Transportation Cost because a supplier may have a high total cost simply because it handles many shipments, while another supplier may have a high average cost per shipment.

6.3 Average Delivery Days

Average Delivery Days = AVERAGE(FactTransportation[Delivery_Days])

Purpose

Calculates the average transportation delivery duration.This is the main time-performance measure for the transportation layer.

It is different from Milestone 2's **Average Supplier Lead Time**, which measures supplier delivery-related time in the original supply-chain data.

6.4 Total Distance

```
Total Distance =
SUM(FactTransportation[Distance_KM])
```

Purpose

Calculates the total transportation distance in kilometres.

It provides a measure of the physical transportation footprint.

6.5 Transportation Cost per KM

The following measure was created:

```
Cost per KM New =
DIVIDE(
    [Total Transportation Cost],
    SUM(FactTransportation[Distance_KM]),
    0
)
```

Purpose

This is a **distance-normalized transportation efficiency measure**.

Raw transportation cost alone can be misleading because longer-distance shipments naturally tend to cost more.

Cost per KM makes it easier to compare:

- Suppliers
- Routes
- Carriers
- Transportation modes

The DIVIDE() function also safely handles situations where total distance is zero.

7. Advanced KPI Visualizations

Five KPI cards were created:

KPI Card	Measure	Purpose
Total Transportation Cost	Total Transportation Cost	Overall transportation expenditure
Average Transportation Cost	Average Transportation Cost	Typical transportation cost
Average Delivery Days	Average Delivery Days	Average delivery duration
Total Transportation Distance (KM)	Total Distance	Total distance covered
Transportation Cost per KM	Cost per KM New	Transportation cost efficiency

KPI Card Formatting

The five cards use a consistent dashboard design:

- Dark-gray background: **#252B30**
- Large and bold KPI values
- Smaller category labels
- Cards arranged in one horizontal row
- Consistent visual formatting

The cards respond dynamically to the dashboard slicers.

8. Supplier Performance Analytics

8.1 Transportation Cost by Supplier

A **Clustered Column Chart** was created.

Property	Configuration
Visual	Clustered Column Chart
X-axis	Supplier_ID
Y-axis	Total Transportation Cost
Sort	Descending
Title	Transportation Cost by Supplier

Purpose

This chart compares transportation expenditure associated with each supplier. Sorting in descending order places the highest-cost suppliers first, making them easier to investigate for:

- Cost reduction
- Rate negotiation
- Shipment consolidation
- Supplier performance review

9. Average Delivery Days by Supplier

A **Clustered Bar Chart** was created.

Property	Configuration
Visual	Clustered Bar Chart
Y-axis	Supplier_ID
X-axis	Average Delivery Days
Sort	Descending
Title	Average Delivery Days by Supplier

Purpose

This chart compares average transportation delivery duration between suppliers.

When combined with transportation cost analysis, it helps identify:

- Suppliers that are expensive and slow.
- Suppliers that are cheap but slow.
- Suppliers that provide faster delivery at higher cost.

This provides a better supplier-performance view than cost analysis alone.

10. Route Performance Analysis

10.1 Transportation Cost by Route

A **Pie Chart** was created.

Property	Configuration
Visual	Pie Chart
Legend	Route_ID
Values	Total Transportation Cost
Title	Transportation Cost by Route

Purpose

The chart shows the relative share of transportation cost associated with each route. It helps identify whether:

- Transportation expenditure is evenly distributed.
- A small number of routes account for a large portion of the total cost.
- Certain routes may require further investigation.

11. Carrier Performance Analysis

11.1 Transportation Cost by Carrier

A **Donut Chart** was created.

Property	Configuration
Visual	Donut Chart
Legend	Carrier_ID
Values	Total Transportation Cost
Title	Transportation Cost by Carrier

Purpose

The donut chart provides a part-to-whole view of transportation expenditure by carrier. It helps identify whether the organization is heavily dependent on one carrier. High dependency on one carrier can become a supply-chain resilience risk even when that carrier currently offers competitive rates.

12. Transportation Mode Analysis

12.1 Transportation Cost by Mode

A **Treemap** was created.

Property	Configuration
Visual	Treemap
Category	Transport_Mode
Values	Total Transportation Cost
Title	Transportation Cost by Mode

Purpose

The treemap compares transportation costs across modes such as:

- Road
- Rail
- Air

It makes differences in transportation expenditure easy to identify visually.

13. Distance and Transportation Cost Analysis

13.1 Distance vs Transportation Cost

A **Scatter Chart** was created.

Property	Configuration
Visual	Scatter Chart
X-axis	Distance_KM
Y-axis	Transportation_Cost
Legend	Carrier_ID
Title	Distance vs Transportation Cost

Purpose

The scatter chart examines the relationship between transportation distance and transportation cost. Normally, transportation cost increases as distance increases. The important part of the analysis is identifying **outliers** — shipments that have unusually high costs compared with their distance. The Carrier_ID legend also helps identify whether a particular carrier consistently appears above the general cost-distance pattern.

14. Delivery Performance by Transportation Mode

14.1 Average Delivery Days by Transportation Mode

A **Line Chart** was created.

Property	Configuration
Visual	Line Chart
X-axis	Transport_Mode
Y-axis	Average Delivery Days
Title	Average Delivery Days by Transportation Mode

Purpose

This visual compares delivery duration across transportation modes. When viewed together with transportation cost by mode, it provides a **cost-versus-speed comparison**. For example, the analysis can help understand the trade-off between:

- Faster but more expensive transportation
- Slower but lower-cost transportation

15. Interactive Slicers

Four slicers were added:

Slicer	Field	Purpose
Year Month	Year Month	Filter by period
Supplier	Supplier_ID	Filter by supplier
Warehouse	Warehouse_ID	Filter by warehouse
Transportation Mode	Transport_Mode	Filter by mode

The slicers were placed together at the top of the dashboard. Selecting a value updates the KPI cards and connected transportation visuals according to the selected filter context.

16. Supplier & Transportation Analytics Dashboard

The complete dashboard contains **16 components**:

No.	Dashboard Component	Analysis
1	Total Transportation Cost	Overall transportation expenditure
2	Average Transportation Cost	Average cost per record
3	Average Delivery Days	Average delivery duration
4	Total Transportation Distance (KM)	Total distance
5	Transportation Cost per KM	Cost efficiency
6	Transportation Cost by Supplier	Supplier-level cost
7	Average Delivery Days by Supplier	Supplier delivery comparison
8	Transportation Cost by Route	Route cost distribution
9	Transportation Cost by Carrier	Carrier cost distribution
10	Transportation Cost by Mode	Mode cost comparison
11	Distance vs Transportation Cost	Distance-cost relationship
12	Average Delivery Days by Transportation Mode	Mode delivery comparison
13	Year Month Slicer	Period filtering
14	Supplier_ID Slicer	Supplier filtering
15	Warehouse_ID Slicer	Warehouse filtering
16	Transport_Mode Slicer	Mode filtering

17. Supplier & Transportation Analytical Flow

The Milestone 3 analytical process follows this sequence:

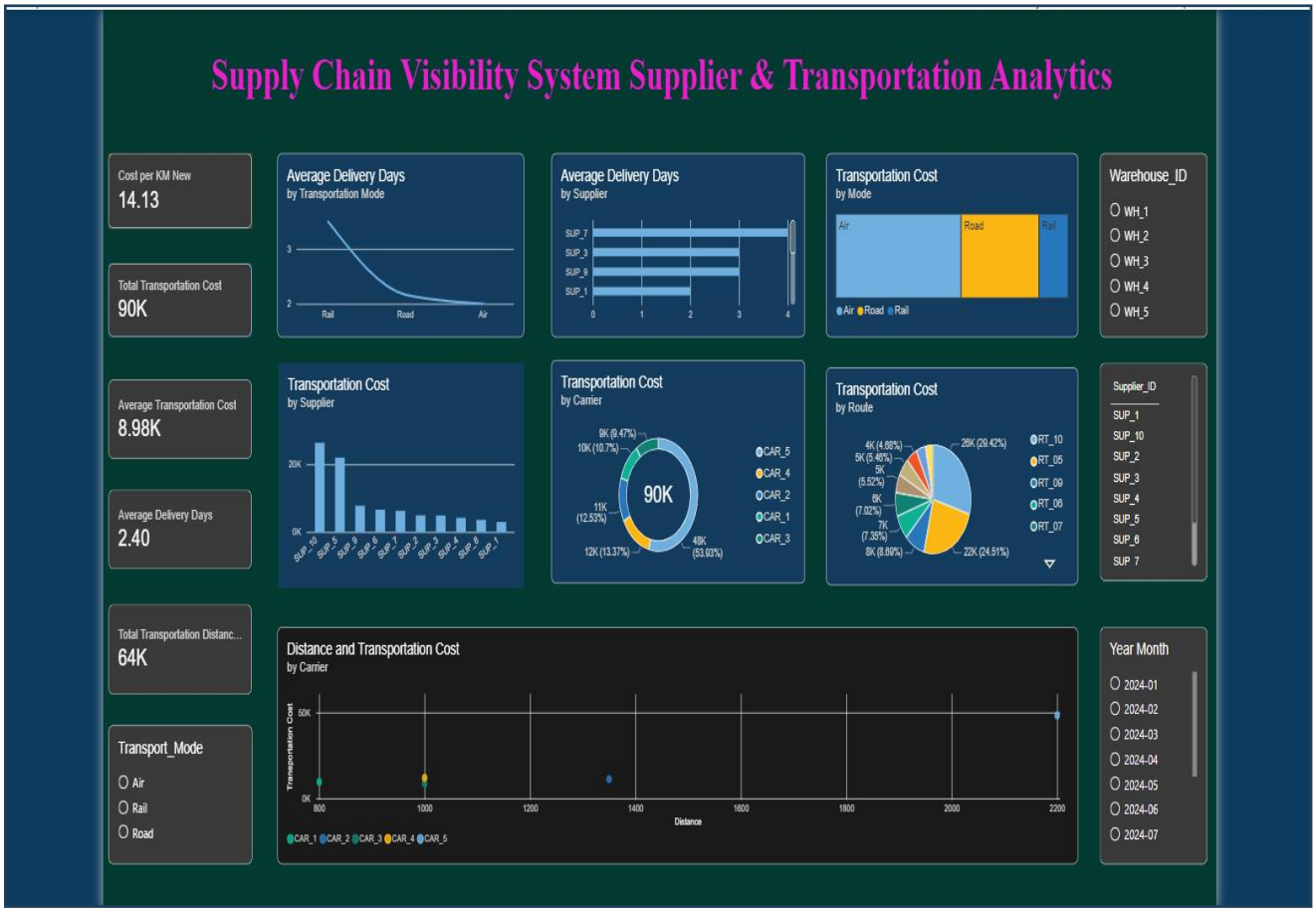
Supply Chain Data + Transportation Data
↓
Transportation Data Modelling
↓
Transportation DAX Measures
↓
Supplier Cost & Delivery Analysis
↓
Route & Carrier Analysis
↓
Transportation Mode Analysis
↓
Distance & Cost Efficiency Analysis
↓
Interactive Slicers
↓
Supplier & Transportation Analytics Dashboard

The dashboard enables the following business analysis:

- Compare supplier transportation cost with delivery duration.
- Identify expensive but fast suppliers.
- Identify cheap but slow suppliers.
- Review route-level transportation cost distribution.
- Identify concentration of transportation expenditure.
- Monitor carrier-level transportation expenditure.
- Identify dependency on a single carrier.
- Compare transportation modes using cost and delivery duration.
- Identify shipments with unusually high cost compared with distance.
- Use **Cost per KM** as a normalized transportation-efficiency measure.
- Filter the complete dashboard by period, supplier, warehouse and transportation mode.

21. DAX Measures Created

Measure	Formula
Total Transportation Cost	SUM(FactTransportation[Transportation_Cost])
Average Transportation Cost	AVERAGE(FactTransportation[Transportation_Cost])
Average Delivery Days	AVERAGE(FactTransportation[Delivery_Days])
Total Distance	SUM(FactTransportation[Distance_KM])
Cost per KM New	DIVIDE([Total Transportation Cost], SUM(FactTransportation[Distance_KM]), 0)



Milestone 3 extends the Supply Chain Visibility System into **Supplier & Transportation Analytics**. A separate transportation fact table was introduced to analyze transportation cost, delivery duration, distance, routes, carriers and transportation modes without affecting the existing Milestone 1 and Milestone 2 model.

The dashboard combines **five advanced KPI cards, seven analytical visuals and four interactive slicers** to provide a complete view of supplier and transportation performance.

MILESTONE 4

Dashboard Performance & Optimization Analytics - Warehouse Efficiency Report | Executive Overview | Performance Optimization | Advanced KPIs | Interactive Dashboards

1. Introduction

Milestone 4 focuses on bringing the analytical work developed in the previous milestones into **three specialized dashboards**.

The objective is to provide different levels of analysis for different users:

1. **Warehouse Efficiency Report** – focuses on warehouse-level operational performance.
2. **Executive Overview** – provides a high-level summary of overall business performance.
3. **Dashboard Performance Optimization** – focuses on transportation efficiency, delivery performance and inventory optimization.

This milestone converts the detailed analysis from earlier milestones into focused dashboards that can be used for **operational monitoring, management review and optimization analysis**.

2. Problem Statement

The main objectives of Milestone 4 are:

- Analyze warehouse efficiency.
- Provide an executive-level overview of the supply chain.
- Monitor important business KPIs.
- Analyze transportation performance.
- Compare warehouse profitability and inventory.
- Identify areas for operational improvement.
- Create interactive dashboards using slicers.
- Present the project results in a clear and management-friendly format.

3. Dashboard Overview

Three dashboards were developed:

Dashboard	Main Purpose
Warehouse Efficiency Report	Analyze warehouse inventory, sales, orders and profitability
Executive Overview	Provide a high-level view of business performance
Dashboard Performance Optimization	Analyze transportation, delivery and inventory efficiency

4. Warehouse Efficiency Report

4.1 Purpose

The **Warehouse Efficiency Report** focuses on understanding how individual warehouses are performing.

It helps answer questions such as:

- How much inventory is held by each warehouse?
- How many units are sold?
- How much profit does each warehouse generate?
- Which warehouse has higher inventory?
- How are sales distributed across warehouses?
- How does inventory compare with warehouse performance?

5. Warehouse Efficiency KPI Cards

Four KPI cards were created.

5.1 Total Order Quantity

Shows the total quantity of orders handled by the selected warehouse or filter context.

5.2 Total Inventory Value

Shows the value of inventory associated with the selected warehouse.

5.3 Total Units Sold

Shows the total quantity of units sold.

5.4 Average Inventory Level

Shows the average inventory level.

These KPIs provide a quick summary of warehouse activity before users analyze the detailed charts.

6. Warehouse Efficiency Visualizations

6.1 Average Inventory Level by Warehouse

Visual: Clustered Column Chart

- X-axis: Warehouse
- Y-axis: Average Inventory Level

Purpose

This visual compares the average inventory level maintained by different warehouses. It helps identify warehouses with relatively high or low inventory levels.

6.2 Profit by Warehouse

Visual: Line Chart

- X-axis: Warehouse
- Y-axis: Profit

Purpose

This visual compares profitability between warehouses. It helps identify warehouses generating higher or lower profit and provides a financial view of warehouse performance.

6.3 Sales Distribution by Warehouse

Visual: Donut Chart

- Legend: Warehouse
- Values: Total Sales

Purpose

This visual shows how total sales are distributed among the warehouses. It helps identify which warehouses contribute a larger share of overall sales.

6.4 Inventory Value by Warehouse

Visual: Treemap

- Category: Warehouse
- Values: Total Inventory Value

Purpose

The treemap provides a visual comparison of inventory value across warehouses. Larger areas represent warehouses with higher inventory value.

6.5 Warehouse Performance Summary

Visual: Matrix

The matrix provides a detailed warehouse-level summary. It allows multiple performance measures to be viewed together instead of analyzing each chart separately.

6.6 Supplier Lead Time vs Profit

Visual: Scatter Chart

- X-axis: Supplier Lead Time
- Y-axis: Profit

Purpose

This visual helps explore the relationship between supplier delivery lead time and profitability. It can help identify whether warehouses or related operations associated with longer supplier lead times also show different profit levels.

6.7 Profit Margin % by Warehouse

Visual: Ribbon Chart

- Category: Warehouse
- Values: Profit Margin %

Purpose

This visual compares warehouse profitability using **Profit Margin %**, rather than only looking at absolute profit. This helps compare warehouses more fairly when their sales volumes are different.

6.8 Order Quantity by Stock Status

Visual: Pie Chart

Purpose

This visual compares order quantity across the available stock-status categories. It helps understand how order quantities are distributed according to stock condition.

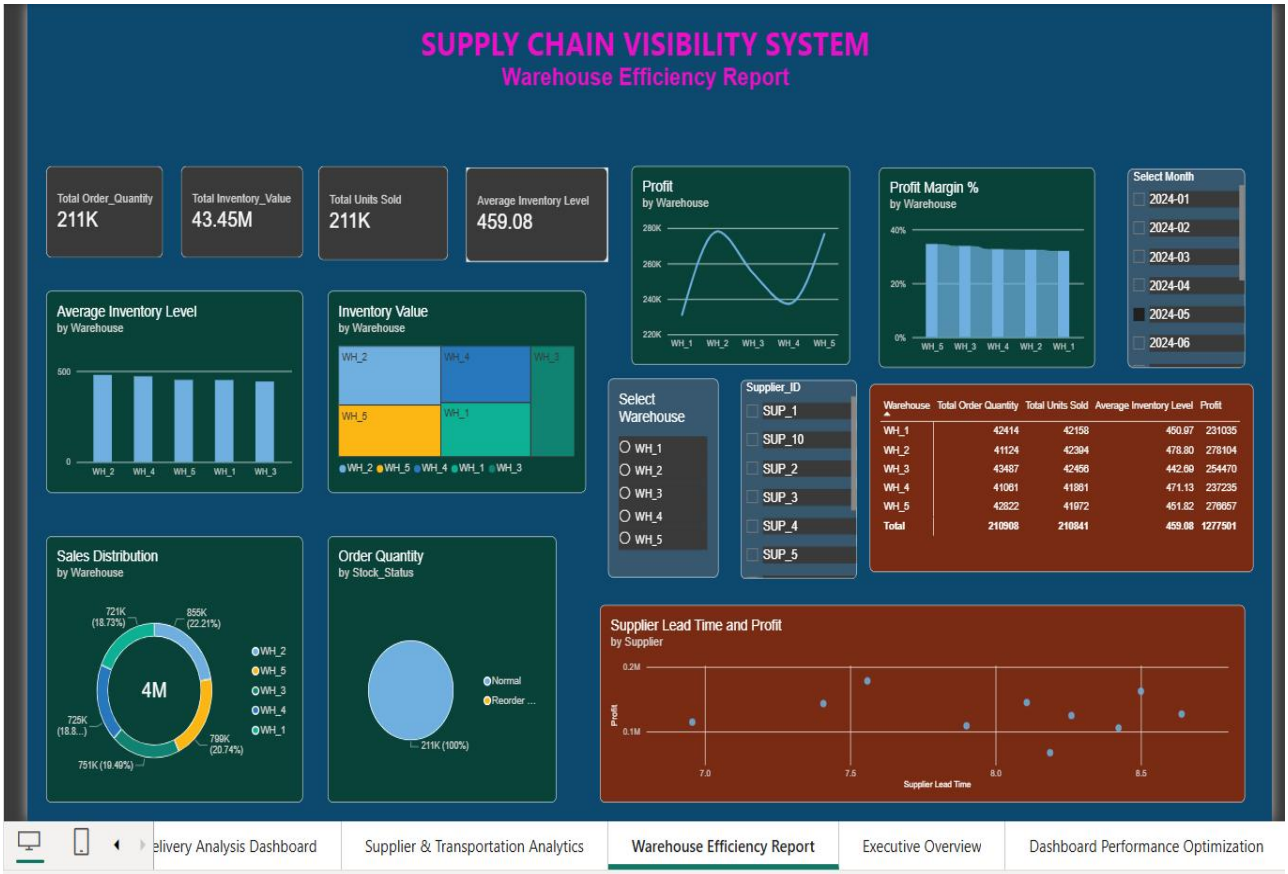
7. Warehouse Efficiency Slicers

Three slicers were added:

Slicer	Purpose
Warehouse	Analyze a specific warehouse
Month	Analyze a selected month or period
Supplier	Analyze supplier-related warehouse performance

These slicers make the Warehouse Efficiency Report interactive.

Warehouse Efficiency Report Dashboard



8. Executive Overview

8.1 Purpose

The **Executive Overview** provides a high-level summary of the entire Supply Chain Visibility System. Instead of showing every detailed operational metric, this dashboard focuses on the most important business indicators.

It is designed to answer:

- What are the total sales?
- How much profit is generated?
- What is the profit margin?
- What is the transportation cost?
- What is the average supplier lead time?
- How are sales changing over time?
- Which warehouses are more profitable?
- How is inventory distributed?

9. Executive KPI Cards

Five KPI cards were created.

9.1 Total Sales

Shows the total sales generated.

9.2 Total Profit

Shows the difference between total sales and total COGS.

9.3 Profit Margin %

Shows profitability relative to sales.

9.4 Total Transportation Cost

Shows total transportation expenditure.

9.5 Average Supplier Lead Time

Shows the average supplier lead time.

Together, these five KPIs provide a quick overview of the major financial, operational and logistics indicators.

10. Executive Overview Visualizations

10.1 Monthly Sales Trend

Visual: Line Chart

- X-axis: Year Month
- Y-axis: Total Sales

Purpose

Shows how sales change over time. It helps management identify periods of higher or lower sales activity.

10.2 Profit by Warehouse

Visual: Bar Chart

- Category: Warehouse
- Values: Total Profit

- Sorted in descending order

Purpose

Compares warehouse profitability and makes the highest- and lowest-performing warehouses easy to identify.

10.3 Transportation Cost by Mode

Visual: Treemap

- Category: Transport Mode
- Values: Total Transportation Cost

Purpose

Shows the contribution of each transportation mode to total transportation expenditure. This helps management understand the transportation-cost structure.

10.4 Inventory Value by Warehouse

Visual: Column Chart

- X-axis: Warehouse
- Y-axis: Total Inventory Value
- Sorted in descending order

Purpose

Compares the value of inventory held by each warehouse. This helps management understand where inventory value is concentrated.

10.5 Profit Margin % by Warehouse

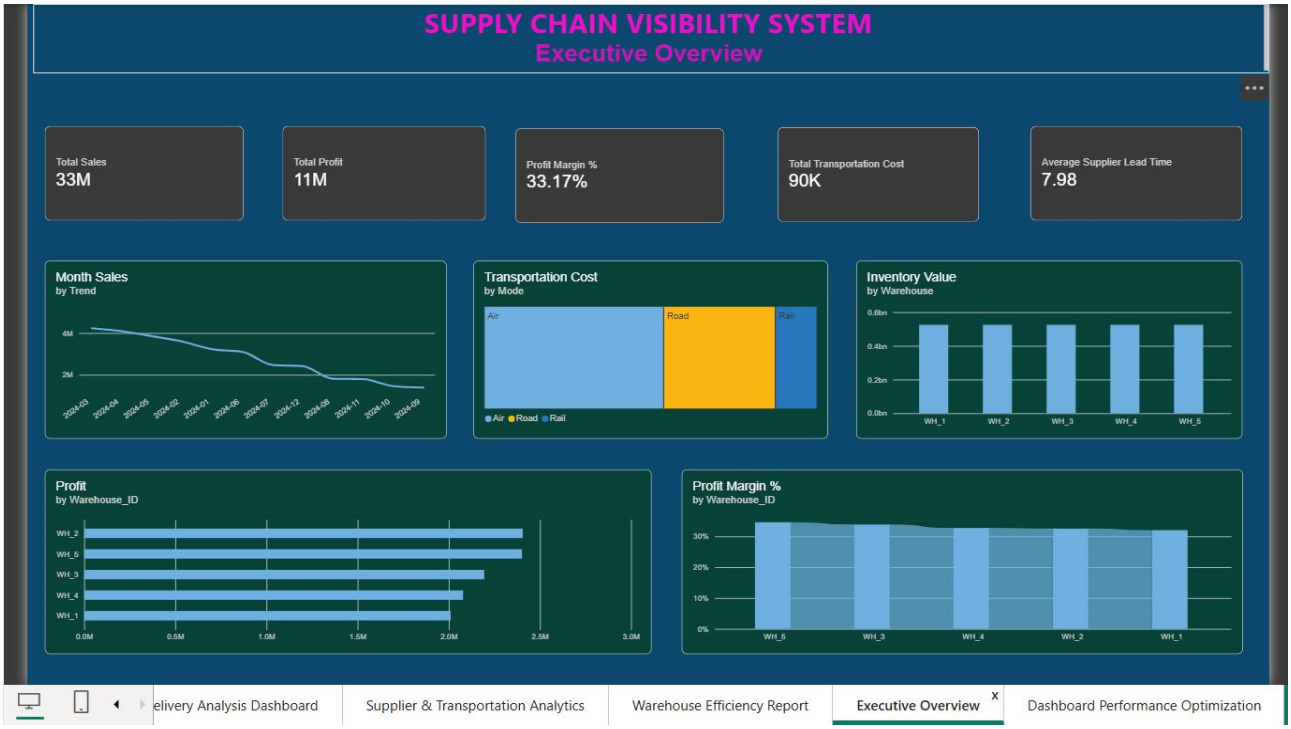
Visual: Ribbon Chart

- Category: Warehouse
- Values: Profit Margin %

Purpose

Compares warehouse profitability based on percentage margin. This provides a more normalized view of warehouse performance.

Executive Overview Dashboard



11. Dashboard Performance Optimization

11.1 Purpose

The **Dashboard Performance Optimization** page focuses on identifying opportunities to improve transportation efficiency, delivery performance and inventory management.

It combines transportation metrics from Milestone 3 with warehouse inventory analysis.

The dashboard helps answer:

- How much does transportation cost per kilometre?
- How long does transportation take?
- Which suppliers have higher transportation costs?
- Which routes are more expensive?
- Which warehouses have higher average inventory?
- How does transportation cost relate to distance?

12. Optimization KPI Cards

Four KPI cards were created.

12.1 Transportation Cost per KM

Measures transportation cost relative to distance travelled.

12.2 Average Delivery Days

Shows the average transportation delivery duration.

12.3 Average Inventory Level

Shows the average inventory maintained.

12.4 Total Transportation Distance

Shows the total transportation distance in kilometres.

These KPIs provide a quick view of transportation and inventory efficiency.

13. Performance Optimization Visualizations

13.1 Transportation Cost vs Distance

Visual: Scatter Chart

- X-axis: Distance_KM
- Y-axis: Transportation_Cost
- Legend: Carrier_ID

Purpose

Shows the relationship between transportation distance and transportation cost. It helps identify unusually expensive transportation records relative to their distance.

13.2 Average Delivery Days by Transportation Mode

Visual: Line Chart

- X-axis: Transport_Mode
- Y-axis: Average Delivery Days

Purpose

Compares delivery duration across transportation modes. This helps understand the relationship between transportation mode and delivery speed.

13.3 Transportation Cost by Supplier

Visual: Column Chart

- X-axis: Supplier_ID
- Y-axis: Total Transportation Cost

- Sorted descending

Purpose

Identifies suppliers associated with higher transportation expenditure. This can support supplier-cost review and transportation-cost optimization.

13.4 Average Inventory Level by Warehouse

Visual: Bar Chart

- Y-axis: Warehouse
- X-axis: Average Inventory Level
- Sorted descending

Purpose

Compares inventory levels between warehouses. This can help identify warehouses carrying relatively high levels of inventory.

13.5 Transportation Cost by Route

Visual: Donut Chart

- Legend: Route_ID
- Values: Total Transportation Cost

Purpose

Shows how transportation expenditure is distributed among routes. This helps identify routes that contribute significantly to transportation costs.

14. Interactive Slicers

The Dashboard Performance Optimization page contains:

Slicer	Purpose
Year Month	Filter analysis by time
Supplier_ID	Filter supplier performance
Warehouse_ID	Filter warehouse performance
Transport_Mode	Filter transportation mode

The slicers allow users to analyze the optimization dashboard from different perspectives.

15. Dashboard Design

A consistent design was applied across the three Milestone 4 dashboards.

Main Heading

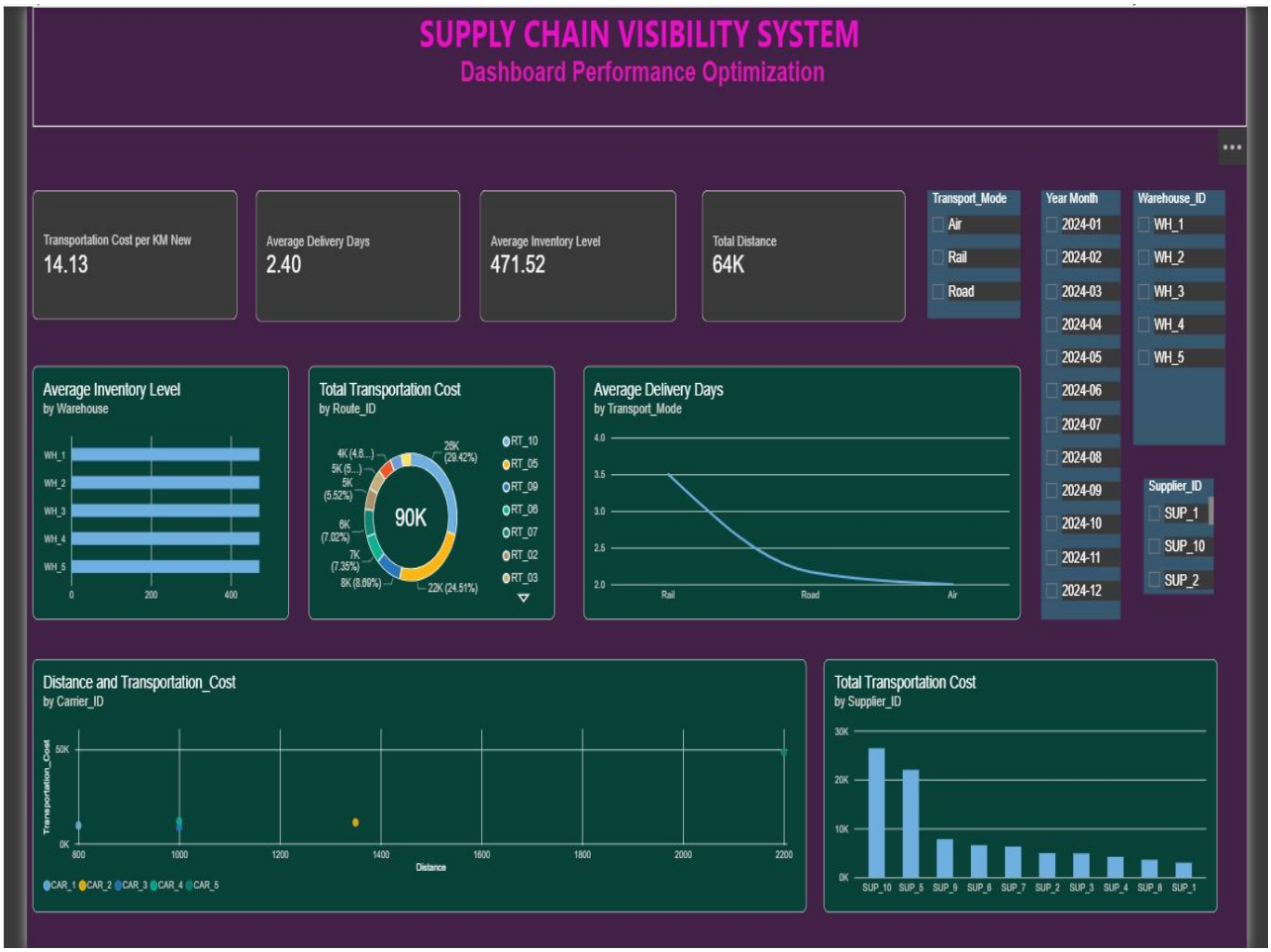
SUPPLY CHAIN VISIBILITY SYSTEM

Dashboard Subtitles

- **Warehouse Efficiency Report**
- **Executive Overview**
- **Dashboard Performance Optimization**

The dashboards use a consistent visual style so that users can move between pages without having to learn a different layout.

Dashboard Performance Optimization



16. Analytical Flow of Milestone 4

The overall analytical flow is:

Supply Chain Data
↓
Existing Data Model & DAX Measures
↓
Warehouse Analysis
↓
Executive-Level KPI Analysis
↓
Transportation & Inventory Optimization
↓
Interactive Slicers
↓
Three Specialized Dashboards
↓
Business Decision Support

Operational Level

The **Warehouse Efficiency Report** helps warehouse teams monitor:

- Inventory
- Orders
- Units sold
- Sales
- Profit
- Profit margin

Management Level

The **Executive Overview** provides management with:

- Sales
- Profit
- Profit margin
- Transportation cost
- Supplier lead time
- Sales trends
- Warehouse performance

Optimization Level

The **Dashboard Performance Optimization** helps identify:

- Transportation cost inefficiencies
- High-cost suppliers
- Expensive routes
- Transportation-distance relationships
- Delivery-time differences
- High inventory warehouses

The three dashboards make it easier to identify important patterns.

Warehouse Efficiency

Warehouses can be compared using inventory, sales, profit and profit margin instead of relying on a single KPI.

Executive Performance

Management can view the overall business position using a small set of important KPIs.

Transportation Efficiency

Transportation cost can be analyzed together with distance and delivery duration, giving a more complete view than transportation cost alone.

Inventory Optimization

Average inventory levels can be compared across warehouses to identify locations that may require further inventory investigation.

Supplier Analysis

Supplier transportation cost and lead time can be compared to support supplier-performance discussions.

Milestone 4 brings together the analytical work developed throughout the project into three specialized Power BI dashboards.

The **Warehouse Efficiency Report** provides detailed warehouse-level analysis, the **Executive Overview** provides a management-level summary, and the **Dashboard Performance Optimization** focuses on transportation and inventory efficiency.

Together, these dashboards transform the underlying supply-chain data into an interactive reporting solution that supports **operational monitoring, management review and optimization-oriented decision-making**.

CONCLUSION

The **Supply Chain Visibility System with Optimization Analytics** was developed progressively across four milestones using **Power BI, Power Query, DAX, and Excel-based data**. The project transformed raw supply chain data into an interactive analytical system that supports inventory monitoring, supplier analysis, transportation evaluation, warehouse performance analysis, and management-level decision-making.

Milestone 1 established the foundation of the project by cleaning and transforming the supply chain data, creating a structured **star-schema data model**, developing core DAX measures, and building the initial KPI dashboard. This milestone provided a reliable foundation for analyzing sales, costs, inventory, orders, supplier lead time, and profitability.

Milestone 2 expanded the analysis toward **inventory and delivery performance**. Inventory-versus-demand analysis, inventory gap analysis, supplier lead-time analysis, warehouse comparisons, sales and inventory trends, and variance analysis were implemented. These analyses helped provide a more detailed view of inventory availability, supplier delivery performance, and warehouse-level business performance.

Milestone 3 extended the system into **supplier and transportation analytics**. Supplier performance, transportation cost, delivery days, distance, route, carrier, and transportation-mode analysis were incorporated through a separate transportation fact table and shared dimensions. Advanced KPI cards and interactive visualizations were developed to analyze transportation efficiency and supplier performance.

Milestone 4 brought the previous analytical components together into three major dashboards: **Warehouse Efficiency Report, Executive Overview, and Dashboard Performance Optimization**. These dashboards provide different levels of analysis, from warehouse-level operational performance to executive-level KPIs and transportation/inventory optimization. Interactive slicers and visual analytics allow users to explore the data according to different business requirements.

Overall, the project demonstrates how Power BI and DAX can convert supply chain data into an interactive business intelligence solution. The completed system provides a structured approach for monitoring key performance indicators, identifying performance variations, comparing suppliers and warehouses, analyzing transportation costs, and supporting data-driven supply chain management decisions.

REFERENCES

1. Microsoft Learn, **"Power BI Documentation,"** Microsoft.
   Power BI Documentation
2. Microsoft Learn, **"Data Analysis Expressions (DAX) Reference,"** Microsoft.
   DAX Reference
3. Microsoft Learn, **"Learn DAX Basics in Power BI Desktop,"** Microsoft.
   Learn DAX Basics in Power BI Desktop
4. Microsoft Learn, **"Understand Star Schema and the Importance for Power BI,"** Microsoft.
   Star Schema in Power BI
5. Microsoft Learn, **"Tutorial: From Dimensional Model to Stunning Report in Power BI Desktop,"** Microsoft.
   Power BI Dimensional Model Tutorial
6. Microsoft Support, **"What is Excel?"** Microsoft.
   Microsoft Excel
7. Microsoft Support, **"Excel Help & Learning,"** Microsoft.
   Excel Help & Learning
8. Kimball, R., & Ross, M., **The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling**, 3rd Edition, Wiley, 2013. Microsoft also identifies this as a widely adopted reference for dimensional/star-schema modeling.